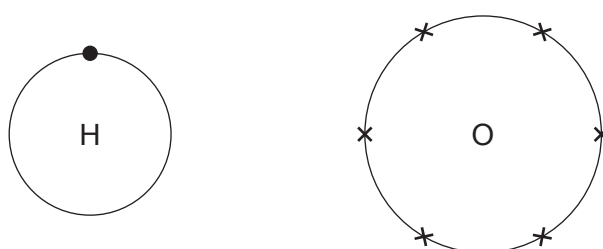


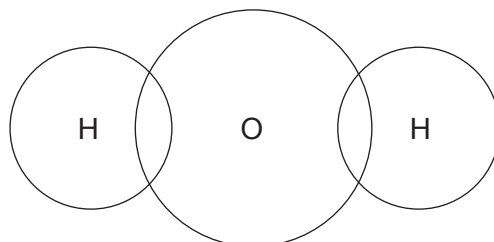
4. (a) The table shows the electronic structure of the elements present in water.

Element	Electronic structure
hydrogen	1
oxygen	2,6

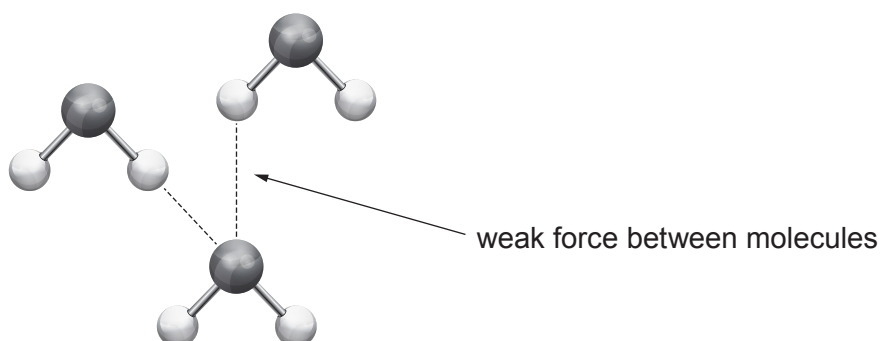
The diagrams show the outer shell electrons in an atom of hydrogen and an atom of oxygen.



- (i) Complete the diagram to show the outer shell electrons in a molecule of water. [2]



- (ii) The diagram shows some water molecules and the weak forces between them.

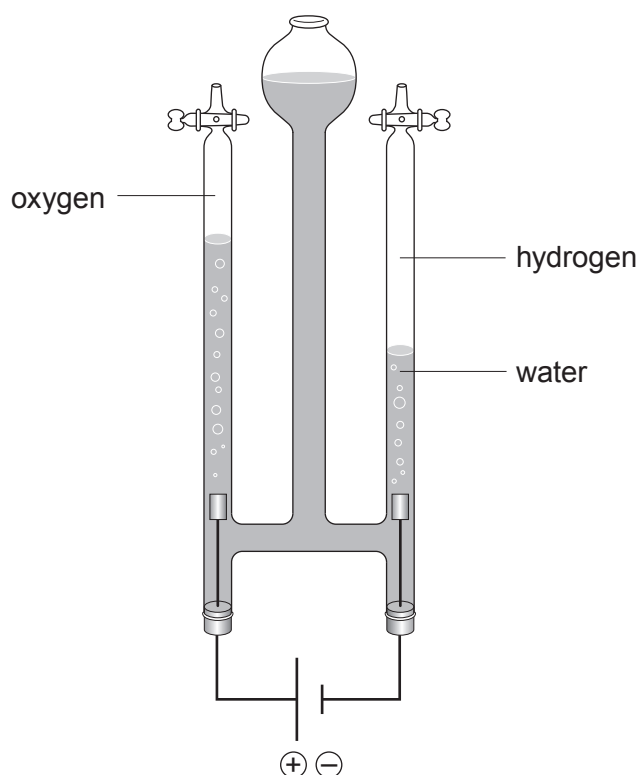


Put a tick (✓) in the box next to the property of water which can be explained by the weak forces between water molecules. [1]

poor conductor of electricity	
colourless	
good conductor of heat	
low melting point and boiling point	



- (b) The diagram shows the apparatus used by a group of students to investigate the volume of hydrogen and oxygen gas formed during the electrolysis of water.



The overall equation for the electrolysis of water is as follows.



- (i) The table shows the total volume of hydrogen formed over 10 minutes.

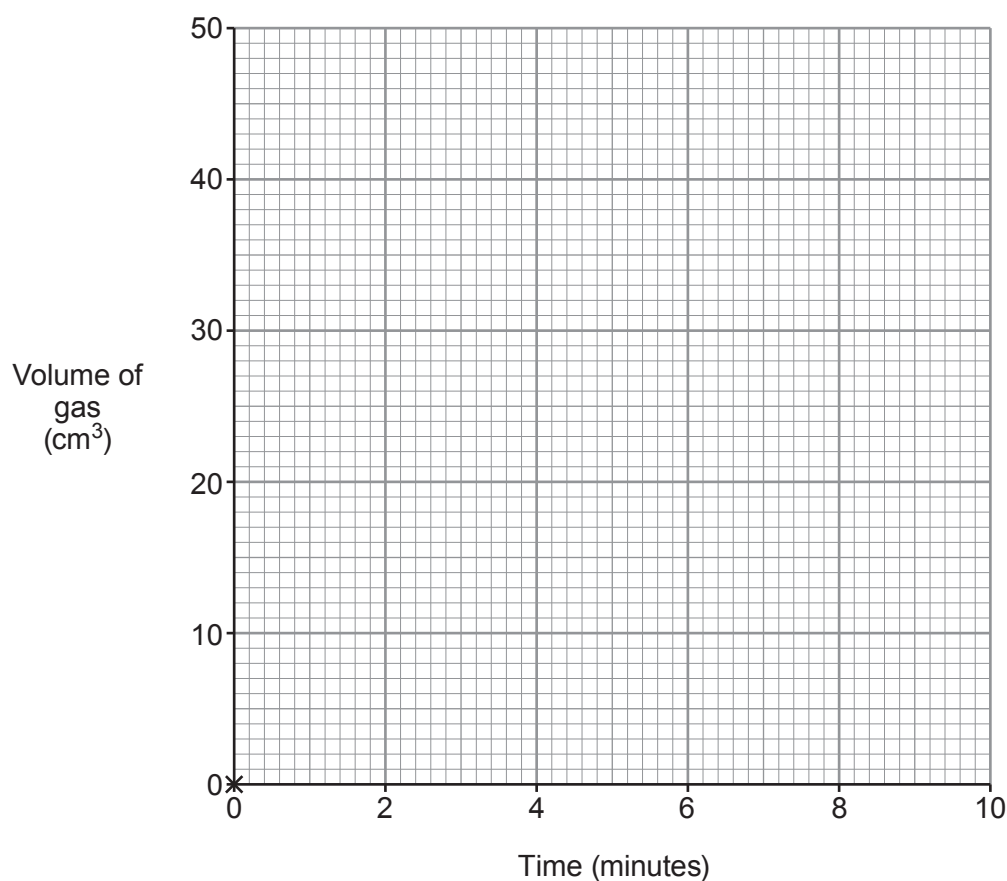
Time (minutes)	0	2	4	6	8	10
Volume of hydrogen (cm ³)	0	10	20	30	40	50



- I. Plot the values from the table on the grid. Draw a suitable line. Label this line '**hydrogen**'.

(0,0) has been plotted for you.

[2]



- II. Using a ruler, draw a second line **on the grid** to show the volume of oxygen that would be collected during the same 10 minutes. Label this line '**oxygen**'. [1]

- III. Describe the relationship between the volume of hydrogen and the volume of oxygen formed during electrolysis. [2]

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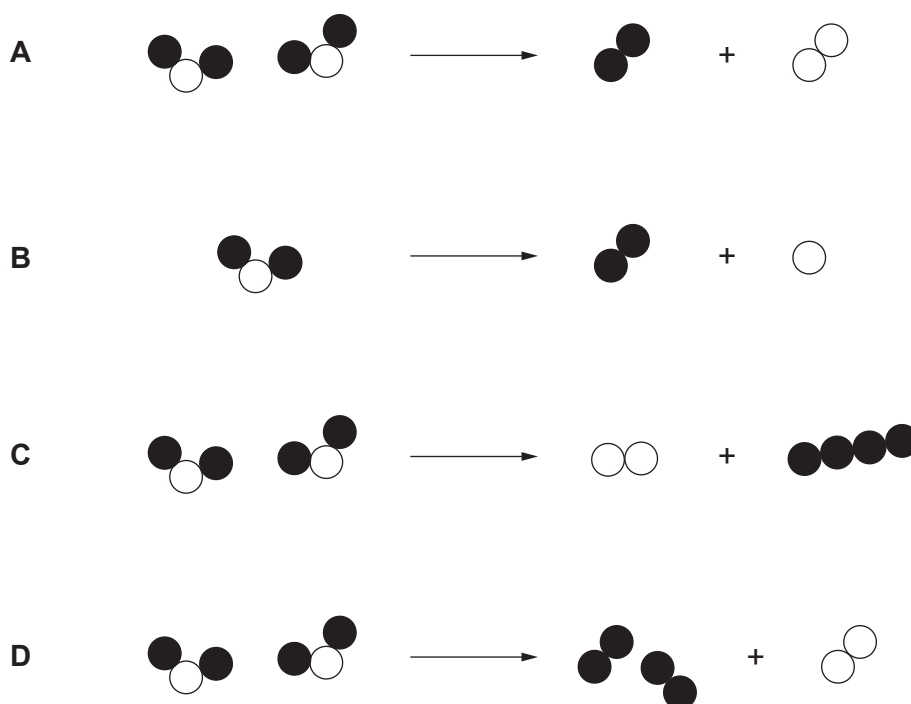


(ii) Electrolysis of water can be represented by the following equation.



Give the **letter, A, B, C or D**, of the diagram which also represents this reaction.

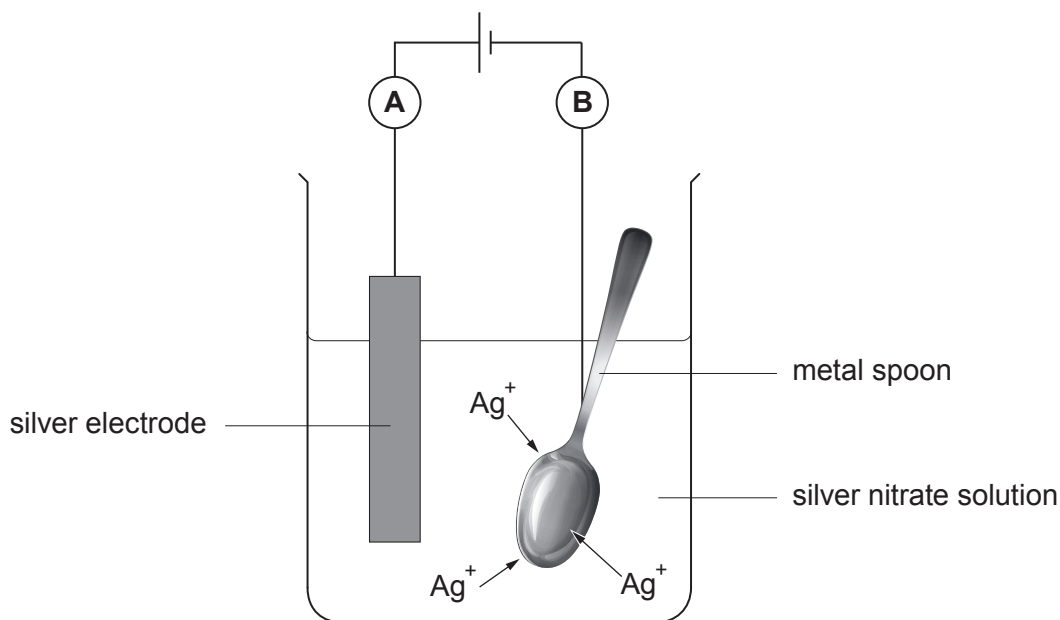
[1]



Letter



- (c) One use of electrolysis is in electroplating. The diagram shows the silver plating of a metal spoon.



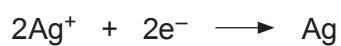
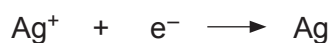
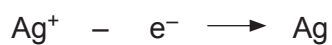
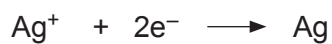
- (i) State why this process does not work for a plastic spoon. [1]

.....

- (ii) Explain why the silver ions move towards the spoon. [2]

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- (iii) Put a tick (✓) in the box next to the electrode equation for the reaction at electrode **B**. [1]


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